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Early Planet Formation in Embedded Disks (eDisk). Structures of molecular outflows

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ABSTRACT

As part of the ALMA Large Program "Early Planet Formation in Embedded Disks" (eDisk), ^{12}CO (2-1) was observed towards 19 nearby low-mass protostars. Of these objects, 15 sources are found to show molecular outflow emission. Based on their morphological and kinematical structures, the CO outflows are classified into three types: a wind-driven shell, where ambient material is swept up by a wide-angle wind from the star, a bow shock, and a slow disk wind, where the molecular gas is launched directly from the disk. We categorize 11 objects as a slow disk wind, 7 as a wind-driven shell, and 1 as a bow shock. Four objects were found to show signs of both slow disk wind and wind-driven shell characteristics. Several objects show significant misalignment between the red- and blue-shifted outflows, and/or between the outflow axis and the minor axis of the dust continuum emission around the protostar. For the objects showing wind-driven shell emission, we compare simple parametrized models with the observations to derive physical properties of the observed shells, such as their dynamical ages. This shows evidence of a time variability in the outflows, such as changes in their direction. In some objects, large differences are seen between the properties of the red- and blue-shifted outflows, possibly indicating differences in the properties of the ambient medium with which the outflow interacts.

Keywords: stars: formation —ISM: jets and outflows

1. INTRODUCTION

Protostellar outflows play an important role in the process of star formation (Shu et al. 1993; Arce & Goodman 2002; Frank et al. 2014; Bally 2016; Pascucci et al. 2023). They enable the circumstellar material to accrete onto the star by extracting angular momentum (e.g. Shu et al. 2000; Bacciotti et al. 2002). They also play a role in the evolution of molecular clouds and cores by injecting energy and momentum into the region surrounding the star (e.g. McKee & Ostriker 2007; Nakamura & Li 2007). Protostellar outflows are made up of several components that can be identified using various types of emission. Jets, which are highly collimated and have high-velocity ($\sim$ a few 100 km s^{-1}), are traced in atomic and ionized emission lines at optical and near-infrared (NIR) wavelengths (e.g. Burrows et al. 1996; Reipurth & Bally 2001). For more embedded sources, they are also traced by molecular lines at millimeter (mm) and sub-millimeter (sub-mm) wavelengths (e.g. Bachiller 1996; Tafalla et al. 2010) and at IR wavelengths (e.g.

McCaughrean et al. 1994; Bally 2016; Ray et al. 2023). Emission from the ionized material at the base of the jet is seen at radio wavelengths (e.g. Anglada 1996; Anglada et al. 1998; Scaife & Heald 2012; Feeney-Johansson et al. 2023). On the other hand, the less collimated component of the outflow is typically slow ($\lesssim$ a few 10 km s^{-1}) compared to the jet and is usually traced by molecular lines such as CO (Snell et al. 1980; Bachiller & Tafalla 1999; Reipurth & Bally 2001; Bally 2016),

Several explanations have been proposed for the origin of the mm/sub-mm CO emission seen in protostellar outflows. The classic and most frequently discussed origin is a bipolar molecular flow of swept-up ambient gas. Such outflows have been seen on scales as large as several parsecs in young stellar objects (YSOs) (e.g Arce et al. 2007; Frank et al. 2014; Bally 2016). The properties of the swept-up gas depend on the driving flow. In the wind-driven shell model (Shu et al. 1991; Lee et al. 2000), a wide-angle wind from the protostar blows into the ambient medium, creating a shock that sweeps the ambient material forming a thin wide-angle shell. Zhang et al. (2019), for example, found multiple shells, which could be formed by a series of outbursts from an intermittent wide-angle wind in the HH 46/47 molecular outflow. In the jet-driven bow shock model (Lee et al. 2000), a collimated high-velocity jet propagates into the ambient medium. When the jet interacts with the ambient medium, bow shocks can arise both as the tips of the terminal shocks or as internal working surfaces, forming a dense shell of shocked molecular gas around the jet. Bow shocks can be produced by variation of velocities or other physical properties; e.g. variations in the mass-loss rate in the jet can produce a series of shocks along the jet axis.

In recent years, sub-mm observations with ALMA have also revealed flows known as slow molecular winds. These are slow ($\lesssim$ a few 10 km s^{-1}) molecular flows observed at scales of $\leq$ 2000 au towards several Class 0 and I objects (Pascucci et al. 2023, and references therein). These flows typically show conical or parabolic shapes with wide opening angles ($10 - 40°$) and an 'onion-like' velocity structure, where the opening angle of the cone is smaller for higher velocities, implying that material further away from the outflow axis is moving at slower velocities. In addition, for some sources (e.g. DG Tau B; de Valon et al. 2020), it has been shown that the flow rotates around the outflow axis in the same direction as the disk. These properties suggest that these flows are tracing material ejected directly from the surface of the Keplerian disk and appear to originate from the disk at radii $\sim 10 - 100$ au. Bjerkeli et al. (2016), for example, observed TMC1A with a spatial resolution of 6 au to find that gas is ejected from a region extending up to a radial distance of 25 au from the central protostar, and that angular momentum is removed from an extended region of the disk.

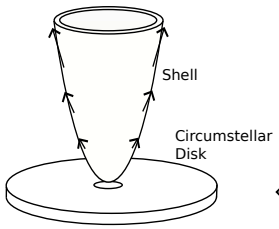
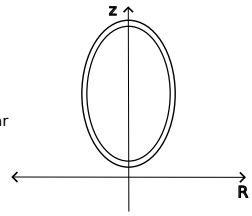
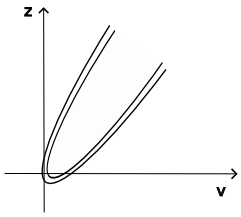
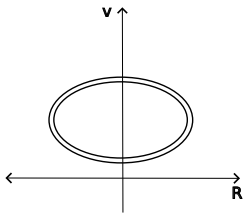
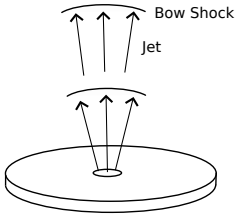
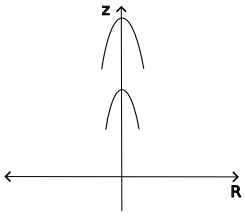
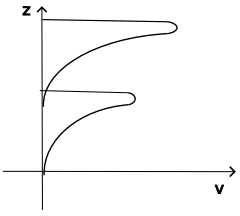
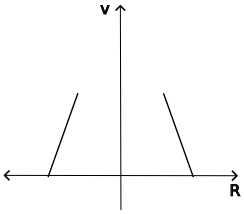
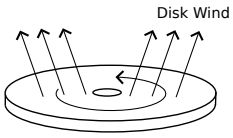
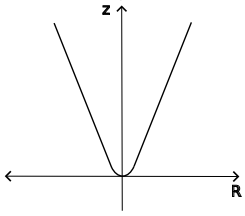
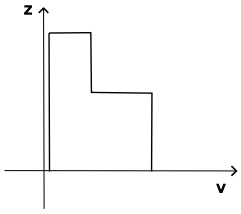
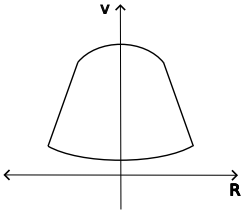
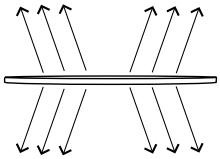
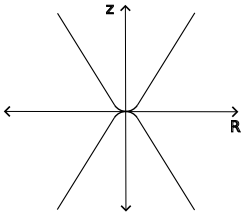
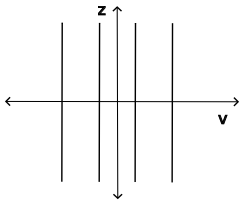
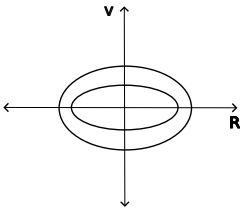
While ALMA has revealed many interesting details about the molecular outflows close to the protostar by high spatial resolution imaging as described above, so far most of them have been on individual objects. Systematic studies of high-resolution molecular outflow emission from a large sample of objects will allow us to measure the fraction of objects that show molecular outflow emission, as well as the frequency with which different types of outflow emission are seen. It will also enable us to measure the frequency of shell structure and detect evidence of variability in mass-loss rate of outflows.

The "Early Planet Formation in Embedded Disks" (eDisk) project (Ohashi et al. 2023) is an ALMA Large Program with the goal of observing substructures, such as rings and gaps, in the disks around young (Class 0/I) protostars in nearby star-forming regions. This involves high-resolution ($\sim$ 7 au) dust continuum observations of 19 nearby Class 0 and Class I sources. The eDisk spectral setting also includes a number of intriguing molecular lines, such as the ^{12}CO ($J = 2 - 1$) line, a well-known tracer of molecular outflows. The targets protostars are listed in Table 1 in the order of increasing bolometric temperature. Of the 19 sources in the survey, 18 have previously been found to be associated with molecular outflows, with an exception of R CrA IRS 5N (IRS 5N hereinafter) (see also §5.2). These outflows should be detectable if they have structures on the scales that eDisk observations are sensitive to (see §2), while large-scale structures may be filtered out. Therefore, eDisk offers an excellent opportunity for a systematic survey of the molecular outflow emission from embedded sources at a high spatial resolution.

The paper is structured as follows. The observations and data reduction process for the eDisk data is described in Section 2. Specific features of CO outflow emission used to categorize the outflows are described in Section 3. We categorize the observed CO emission features and present the observational data of representative objects in Section 4. In Section 5, we discuss time-variability, objects without clear outflows, and the relationship between outflow features and the evolutionary stage of the central protostars.

2. OBSERVATIONS AND DATA REDUCTION

Table 3. Categorization of molecular outflows based on the channel map and the P-V diagrams.

Type	Structure	Morphology in Channel Map	P-V Diagram along outflow axis	P-V Diagram perpendicular to outflow axis
Wind-driven shell[a]				
Jet-driven bow shock[b]				
Slow disk wind ($i \approx 60°$)[c]				
Slow disk wind ($i \approx 90°$)[d]				

[a] Shu et al. (1991); Li & Shu (1996); Lee et al. (2000, 2001); Arce & Goodman (2002); Shang et al. (2006)

[b] Lee et al. (2000, 2001); Cliffe et al. (1996); Arce & Goodman (2002)

[c] Blandford & Payne (1982); Tomisaka (1998); Machida et al. (2008); de Valon et al. (2022)

[d] Hirota et al. (2017); Tabone et al. (2017)

Table 1. Properties of the eDisk sources

Source Name	Class	$T_{\rm bol}$ K	$i_{\rm cont}$ [c] ^{o}	Emission Blue	Emission Red	Previous Jet/Outflow Detections	ref.[g, h]
BHR71 IRS 2	0	39	31	DW[d]	DW	^{12}CO	[2]**[38]**
B335	0	41	37	DW	DW	^{12}CO, IR ([Fe II], HI, H_2)	[1][2][3][4][5][37][39][40]
L1527 IRS	0	41	~ 90	DW	DW	^{12}CO, Radio	[2][3][6][24]**[25]**
IRAS16253-2429	0	42	68	DW	DW	^{12}CO, IR ([Fe II], [Ne II], [Ni II], [H I])	[7][8]**[26]**
IRAS 16544-1604	0	52	73	WD[e]	WD	^{12}CO	[4][9]**[27]**
GSS30 IRS3	0	50	64	DW / WD	WD	^{12}CO, Radio	[10][11][41]
IRAS15398-3359	0	50	51	BS[f]	BS	^{12}CO, IR ([Fe II], [Ne II], [S I], H_2)	[1][4][12][13]**[28]**
R CrA IRS 5N	0	59	65	–	–	Radio?/ H_2? [i]	[14][15]**[29]**
IRAS04166+2706	0	61	47	DW	DW/WD	^{12}CO	[3][4][16]
R CrA IRAS 32[a]	0	64	69	DW / WD	DW/WD	^{12}CO	[4][12]**[30]**
BHR71 IRS 1	0	66	39	DW	DW	^{12}CO	[1][2]**[38]**
Ced 110 IRS 4[b]	0	68	75	–	–	^{12}CO, Radio, IR(H_2)	[2][12][17][18][19]**[31]**[44]
R CrA IRS 7B[b]	I	88	68	–	–	^{12}CO, Radio / H_2?	[12][14][15]**[32]**
IRAS04302+2247	I	88	84	WD?	–	^{12}CO, IR	[3][20]**33]**
IRAS04169+2702	I	163	44	DW	DW?	^{12}CO, Optical ([S II])	[3][21][43]
TMC-1A	I	183	52	DW/WD	DW	^{12}CO, Radio?, IR ([Fe II], H_2)	[1][2][3][23][36]
Oph IRS43[b]	I	193	78	–	–	^{12}CO	[3]**[34]**
L1489 IRS	I	213	71	DW	DW	^{12}CO, Radio	[2][22]**[35]**
Oph IRS63	I	348	47	WD	WD	^{12}CO	[2][3][4][9][42]

[a] R CrA IRAS 32 is a binary. We assume that the outflow is launched from IRAS 32A to quantify the properties of the outflow, e.g. PA.

[b] Binary sources. The inclination is for the primary source.

[c] Inclination angle derived from the dust continuum emission, assuming that it traces a disk; zero inclination means face-on. For most of the objects, the inclination angles are estimated by 2D Gaussian fitting of the dust continuum (Ohashi et al. 2023). Since the disk can be geometrically thick and since tenvelope emission can also contribute, these values are considered a lower limit.

[d] "DW" stands for disk wind.

[e] "WD" stands for wind-driven shell

[f] "BS" stands for bow shock.

[g] [1]Yang et al. (2018), [2]Yıldız et al. (2015), [3]Bontemps et al. (1996), [4]Dunham et al. (2014), [5]Yen et al. (2010), [6]Reipurth et al. (2004), [7]Hsieh et al. (2016), [8]Narang et al. (2024), [9]Fukui (1989), [10]Friesen et al. (2018), [11]Coutens et al. (2019), [12]van Kempen et al. (2009), [13]Yang et al. (2022), [14]Miettinen et al. (2008), [15]Kumar et al. (2011), [16]Wang et al. (2014), [17]Belloche et al. (2006), [18]Lehtinen et al. (2003), [19]Bally et al. (2006), [20]Lucas & Roche (1997), [21]Gomez et al. (1997), [22]Girart et al. (2002), [23]Harsono et al. (2023), [24]Aso et al. (2017), [25]van't Hoff et al. (2023) [26]Aso et al. (2023), [27]Kido et al. (2023), [28]Thieme et al. (2023), [29]Sharma et al. (2023), [30]Encalada et al. (2024), [31]Sai et al. (2023), [32]Ohashi et al. (2023), [33]Lin et al. (2023), [34]Narayanan et al. (2023), [35]Yamato et al. (2023), [36]Bjerkeli et al. (2016), [37]Bjerkeli et al. (2019), [38]Gavino et al. (2024), [39]Federman et al. (2024), [40]Hodapp et al. (2024) [41]Santamaria-Miranda et al. (2024) [42]Flores et al. (2023) [43]Han et al., in prep. [44]Narang et al., in prep.

[h] eDisk first-look papers are in bold font

[i] Radio emission at the position of IRS 5N was detected by Miettinen et al. (2008), however it is possibly gyrosynchrotron emission due to the stellar magnetosphere rather than thermal emission from the ionized jet. IRS 5N is listed as a possible driving source for several molecular hydrogen emission-line objects (MHOs) by Kumar et al. (2011) but it is highly uncertain whether it is the driving source for these.

Table 2. Properties of the ^{12}CO images used

Source	Briggs weighting	*uv* taper kλ	Synthesized Beam ($\theta_{\rm maj} \times \theta_{\rm min}$, PA)	Noise level mJy beam^{-1}
BHR71 IRS 2	2.0	2000	$(0\farcs13 \times 0\farcs12, -1.31^\circ)$	0.8
B335	0.5	1500	$(0\farcs09 \times 0\farcs08, 55.27^\circ)$	1.4
L1527IRS	2.0	2000	$(0\farcs17 \times 0\farcs13, -19.45^\circ)$	0.8
IRAS16253-2429	2.0	2000	$(0\farcs35 \times 0\farcs25, 76.37^\circ)$	1.2
IRAS 16544-1604	2.0	2000	$(0\farcs24 \times 0\farcs18, 78.76^\circ)$	1.0
GSS30 IRS3	2.0	2000	$(0\farcs34 \times 0\farcs24, 74.96^\circ)$	1.0
IRAS15398-3359	2.0	2000	$(0\farcs17 \times 0\farcs15, -79.76^\circ)$	1.1
R Cr A IRS 5N	2.0	2000	$(0\farcs15 \times 0\farcs12, -85.10^\circ)$	1.0
IRAS04166+2706	2.0	2000	$(0\farcs17 \times 0\farcs14, 7.33^\circ)$	1.0
R CrA IRAS 32	2.0	2000	$(0\farcs16 \times 0\farcs12, -85.08^\circ)$	1.0
BHR71 IRS 1	2.0	2000	$(0\farcs15 \times 0\farcs14, -11.07^\circ)$	0.9
Ced 110 IRS 4	2.0	2000	$(0\farcs18 \times 0\farcs13, -21.33^\circ)$	0.9
R CrA IRS7B	2.0	–	$(0\farcs12" \times 0\farcs10, 85.20^\circ)$	0.8
IRAS04302+2247	2.0	2000	$(0\farcs14 \times 0\farcs11, -18.19^\circ)$	1.0
IRAS04169+2702	2.0	2000	$(0\farcs18 \times 0\farcs15, 6.46^\circ)$	1.0
TMC-1A	0.5	2000	$(0\farcs10 \times 0\farcs06, 36.00^\circ)$	1.7
Oph IRS43	2.0	2000	$(0\farcs23 \times 0\farcs18, -63.95^\circ)$	0.8
L1489IRS	2.0	2000	$(0\farcs16 \times 0\farcs12, 12.21^\circ)$	1.0
Oph IRS63	0.5	–	$(0\farcs40 \times 0\farcs28, 78.87^\circ)$	1.5

3. CHARACTERISTIC FEATURES OF OUTFLOWS

Arce & Goodman (2002) schematically described molecular outflow properties predicted by different entrainment models. Inspired by their work, we analyzed channel maps and position-velocity (P-V) diagrams of eDisk targets to see if we can extract some similar features among them. We found that we can categorize our targets roughly into the three types which are described in the following subsections. We interpret them as (i) wind-driven shell, (ii) bow shock, (iii) slow disk wind (Table 3). It should, however, be noted that our categorization is basically qualitative and that we do not aim to validate or exclude specific theoretical models.

3.1. *Wind-driven Shell*

In the velocity channel maps, several of our targets show ellipses or shell-like structures that become larger and shift further away from the protostar at higher velocity channels. We interpret them as wind-driven shells.

When a wide-angle wind from the central protostar blows into ambient material, it sweeps up material into a momentum-conserving shell that is expanding like Hubble law (e.g. Shu et al. 1991; Li & Shu 1996; Lee et al. 2000, 2001; Arce & Goodman 2002; Shang et al. 2006). A cut parallel to the outflow axis results in a parabola on the P-V diagram, whose tilt and opening angle depend on the opening angle and inclination i of the shell. Meanwhile, a cut perpendicular to the outflow axis results in an ellipse on the P-V diagram with the ellipse increasing in size further along the outflow axis.

3.2. *Bow shocks*

When a shell-like structures appears in almost the same position in multiple velocity channels, it means that the shell contains gas of relatively large velocity range. We interpret them as bow shocks.

While bow shocks are ubiquitous in various astrophysical objects, bow shocks caused by the interaction between a fast collimated jet and ambient material are often investigated/modeled in the context of protostellar outflows (e.g. Cliffe et al. 1996; Lee et al. 2000, 2001; Arce & Goodman 2002). A shell structure produced by a jet-driven bow shock is often very elongated along the jet axis (Lee et al. 2001, e.g.), with the largest velocity pointed along the z-direction.In velocity channel maps, the emission appears as an inverted V-shaped curve, with its tip at the position of the bow shock. The P-V diagram along the jet axis shows a spur-like feature, with the velocities increasing rapidly

up to the position of the bow shock, where a broad range of velocities is present. The steepness of the spur-like feature close to the bow shock depends on the inclination angle of the jet.

3.3. *Slow disk wind*

Many of our targets show conical or parabolic flow in the velocity channel maps of $|v| \lesssim 20$ km s^{-1} with respect to the systemic velocity ($v_{\rm sys}$). They also show an onion-like velocity structure; i.e. the opening angle of the conical feature is narrower at higher velocity channels. These features could be interpreted as centrifugal winds whose velocity peaks decrease with distance from the central star.

Theoretical models and numerical simulations predict centrifugal winds launching from a rotating disk (e.g. Blandford & Payne 1982; Shu et al. 1994; Tomisaka 1998; Machida et al. 2008; de Valon et al. 2022), with varying launch radii ranging from the inner disk to outer regions. The overall shape of the wind is conical or parabolic. Since the disk rotation is faster at the inner radii, the wind velocity is higher in the flow originating at the inner disk radii. The actual theoretical models and simulations vary in their respective predictions and features. In our CO emission analysis in this work, we limit the consideration of the slower molecular winds, referred to as those arising from the outer radii. For example, in one such attempt, de Valon et al. (2022) calculated synthetic images by assuming an axisymmetric flow having reached its terminal velocity at the detection location and a constant poloidal velocity along its trajectory.

The emission predicted based on the slow disk wind models highly depends on the inclination. When the disk inclination is $i \approx 90°$, i.e. the axis of the disk wind is on plane of the sky, only low-velocity emission is seen, as we only observe the radial and rotation component of the outflow velocity. In P-V diagrams perpendicular to the outflow axis, the emission traces an ellipse symmetric around the origin ($v = 0, \mathrm{offset} = 0$), if the radial velocity dominates over rotation. If rotation is significant, the ellipse is tilted in the P-V diagram (e.g. Hirota et al. 2017; Tabone et al. 2017).

For outflows inclined to the plane of the sky ($i < 90°$), the velocity maps show a conical or parabolic shape with much higher velocities than the $i = 90°$ case, as we are also seeing the velocity component along the outflow axis. In P-V diagrams perpendicular to the outflow axis, the emission appears as an upside-down cone that is roughly symmetrical around offset= 0. This indicates that the velocity of the emission decreases with distance from the outflow axis, reflecting the onion-like velocity structure with the higher velocity gas located closer to the outflow axis.

Note the emission patterns on the channel maps and the P-V diagrams could overlap for the WD and DW cases at 90∘ for a minimal overlap (if not for any other angle), especially for the base cavity. It is thus important to refer both the channel map and the P-V diagrams to categorize the outflow.

4. RESULTS

The molecular outflow emission detected in 15 of the eDisk sources are divided into the above mentioned 3 types based on the morphology in their channel maps and P-V diagrams. The classifications are summarized in Table 1. Columns 5 and 6 denote the emission type for the blue- and red outflow lobes, respectively, where "DW" stands for slow disk wind, "BS" for bow shock, and "WD" for wind-driven shell. We found that the specific features of wind-driven shell and disk wind coexist in 4 sources which are denoted as "DW/WD". We also note that some sources show emission features quite different between blue and red components regarding dynamical time (see §5.1), opening angle, or coexistence of wind-driven shell and slow disk wind (e.g. TMC-1A).

The channel maps of all the targets are shown in the Appendix A (Figure 10). They are oriented so that the outflow axis is vertical in the panel, with the red-shifted outflow pointing up and blue-shifted outflow pointing down, and the velocity is shifted so that the velocity is relative to the systemic velocity. Unless otherwise stated (see below), the axis of the outflow was determined by assuming that it is perpendicular to the major axis of the continuum emission, which are in turn determined by the Gaussian fits to the dust continuum as discussed in Ohashi et al. (2023) (reference for each source is also shown in bold font in Table 1). We denote this position angle of the major axis of the dust continuum as $\mathrm{PA_{cont}}$ rather than $\mathrm{PA_{disk}}$, since Keplerian rotation is not confirmed for some objects; i.e. the continuum emission may be contaminated by envelope emission. For some sources we note that the outflow axis is obviously not perpendicular to $\mathrm{PA_{cont}}$, and/or the the blue-shifted and red-shifted components have different orientations. We set their outflow axis by finding the axis of symmetry of the cone- or parabola-shaped lobe by eye. The outflow axis for sources with wind-driven shell emission is determined by fitting the shells with a model (see §5 for details). The systemic velocity, the position angle of the outflow axis, and $\mathrm{PA_{cont}}$ for each source are summarized in Table 4. References for the systemic velocity of each object can be found in the first-look papers.

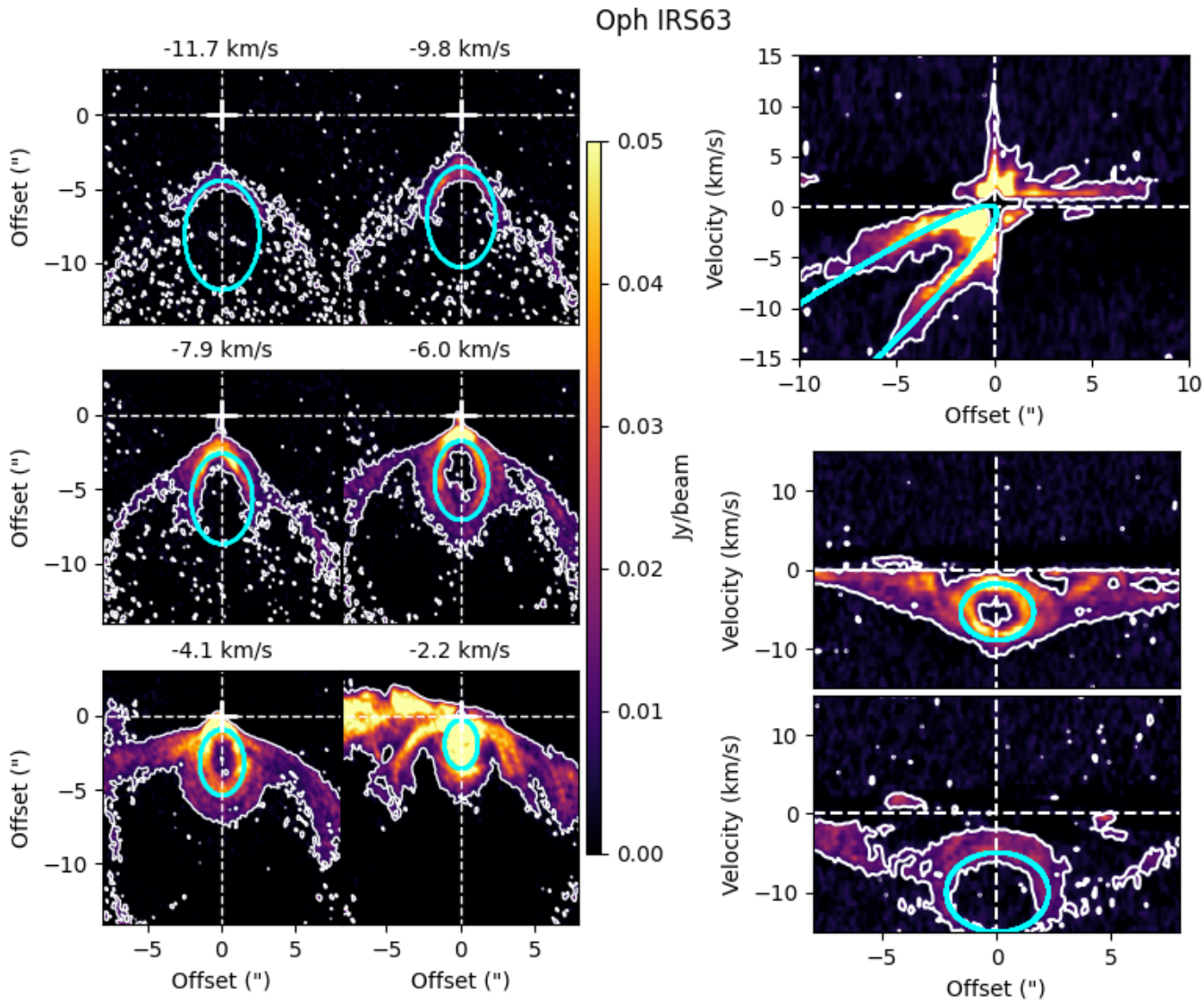


Figure 1. Maps of representative velocity channels (left), position-velocity diagram along the axis of the outflow (top right), and the P-V diagrams perpendicular to the outflow axis at 3″ (middle right) and 6″ from the protostar (bottom right) for Oph IRS 63. The white contours show the 5σ level of the emission, where σ is the root-mean-square noise level given in Table 2. The channel map is rotated so that the outflow axis is in the vertical direction in the panel. The velocity is also shifted to the relative velocity to the systemic velocity. The position angle of the outflow axis and systemic velocity are listed in Table 4. Cyan lines depict the best-fit shell model for shell B1 described in §5.1.

In four objects, IRS 5N, Ced 110 IRS4, R CrA IRS 7B (IRS 7B hereinafter), and Oph IRS43 (hereinafter IRS43), we do not see any clear sign of outflow emission (§5). For one source, IRAS04169+2702 (IRAS04169 hereinafter), we see peculiar features in the red-shifted lobe that cannot be categorized as either of the three types (see §4.5).

We also note that the CO emission can be seen in "molecular jets", where they trace collimated, high-velocity material in the jet, particularly in heavily embedded sources. Among eDisk objects, molecular jets are observed in BHR71 IRS1, BHR71 IRS2, and IRAS04166+2760 (IRAS04166 hereinafter). In the present work, however, we focus on molecular outflows with velocities of $\lesssim 30$ km/s, and do not discuss those molecular jets. For more details, see the eDisk first-look papers for these objects (Gavino et al. (2024); Phuong et al., in prep.)

In this section, we present the channel maps and P-V diagrams of representative objects for emission from a wind-driven shell, a bow shock, a slow disk wind, and a combination of slow disk wind and wind-driven shell.

Table 4. Systemic velocity and PA of the outflow axis

Source	$v_{\rm sys}$ [km/s]	PA[a] [°]	PA method	$\rm PA_{cont}$[b] [°]
BHR71 IRS2	-4.45	152	Sym	68
B335	8.5	275	Sym	163
L1527 IRS	5.9	92	Perp	2
IRAS16253-2429	4.0	203	Perp	113
IRAS 16544-1604[e]	5	128 (B), 335, 340 (R)	Shell	45
GSS30 IRS3	4	162	Shell	109
IRAS15398-3359	5.24	233 (B), 65 (R)	Sym	117
R CrA IRS 5N	6.5	351	Perp	81
IRAS04166+2706	7.0	212	Shell	122
R CrA IRAS 32	5.86	223	Shell	135
BHR71 IRS1	-4.55	359	Sym	98
Ced 110 IRS 4	-4.67	14	Perp	104
R CrA IRS7B	6.0	25	Perp	115
IRAS04302+2247	5.7	265	Perp	175
IRAS04169+2702	6.8	230 (B), 70 (R)	Sym	139
TMC-1A	6.0	166	Shell	76
Oph IRS43	4.0	46	Perp	136
L1489 IRS	7.3	175 (B), 341 (R)	Sym	66
Oph IRS63	2.5	240 (B) 90 (R)	Shell	149

[a] PA= 0 is pointing north and and PA increases counter-clockwise (i.e. east of north). Unless otherwise noted, the value given is the red-shifted PA with the blue-shifted PA being 180° opposite. For objects without clear outflow emission (i.e. R CrA IRS5N, Ced 110 IRS 4, R CrA IRS7B, and Oph IRS43), we adopted the value in this column to draw the channel maps in Figure 10 in Appendix A.

[b] This column indicates the method used to determine the PA of the outflow axis.
Sym: PA is determined by investigating the symmetric axis of the outflow lobes by eye. Two PA values are listed when the axis is different between blue and red lobe.
Perp: PA is taken to be perpendicular to $\rm PA_{cont}$.
Shell: PA is determined from the fitting of the shell model to the velocity channel maps and P-V diagrams (see §5). Two PA values are listed when the axis is different between blue and red lobe.

[c] The PA of the major axis of dust continuum fitted by a 2D Gaussian (Ohashi et al. 2023).

[e] Two PAs are listed for the red-shifted axis as different PA values were derived for the two red-shifted shells (see §5.1.3). The PA of shell 1 was used when creating Figure 8.

4.1. *Wind-driven shell emission*

The source that shows shell structure most clearly is Oph IRS 63. The left panel of Figure 1 shows the maps of Oph IRS 63 at representative blue-shifted velocities. At least one shell-like structure can clearly be seen as an ellipse which shifts further away from the source and becomes larger in higher velocity channels, with emission visible at velocities up to $\approx$ 12 km s^{-1} relative to the systemic velocity (see also Figure 10 in Appendix A). The bottom of the ellipse corresponds to the "front side" of the outflow, while the top of the ellipse corresponds to the "back side" of the outflow at a given velocity channel map. The P-V diagram along the outflow axis clearly shows a parabolic structure in the blue-shifted emission (the top right panel of Fig. 1). The P-V diagrams perpendicular to the outflow axis (bottom right panels) show an elliptical feature, which becomes larger and shifts to higher velocity at larger offsets from the source. All of these properties are exactly what is expected of wind-driven shell emission described in §3.1 and Table 3.

A similar shell-like structure is also seen in the red-shifted outflow of Oph IRS 63. We note that the red-shifted outflow of Oph IRS 63 is much fainter and has a larger opening angle than the blue-shifted outflow (Figure 10 in

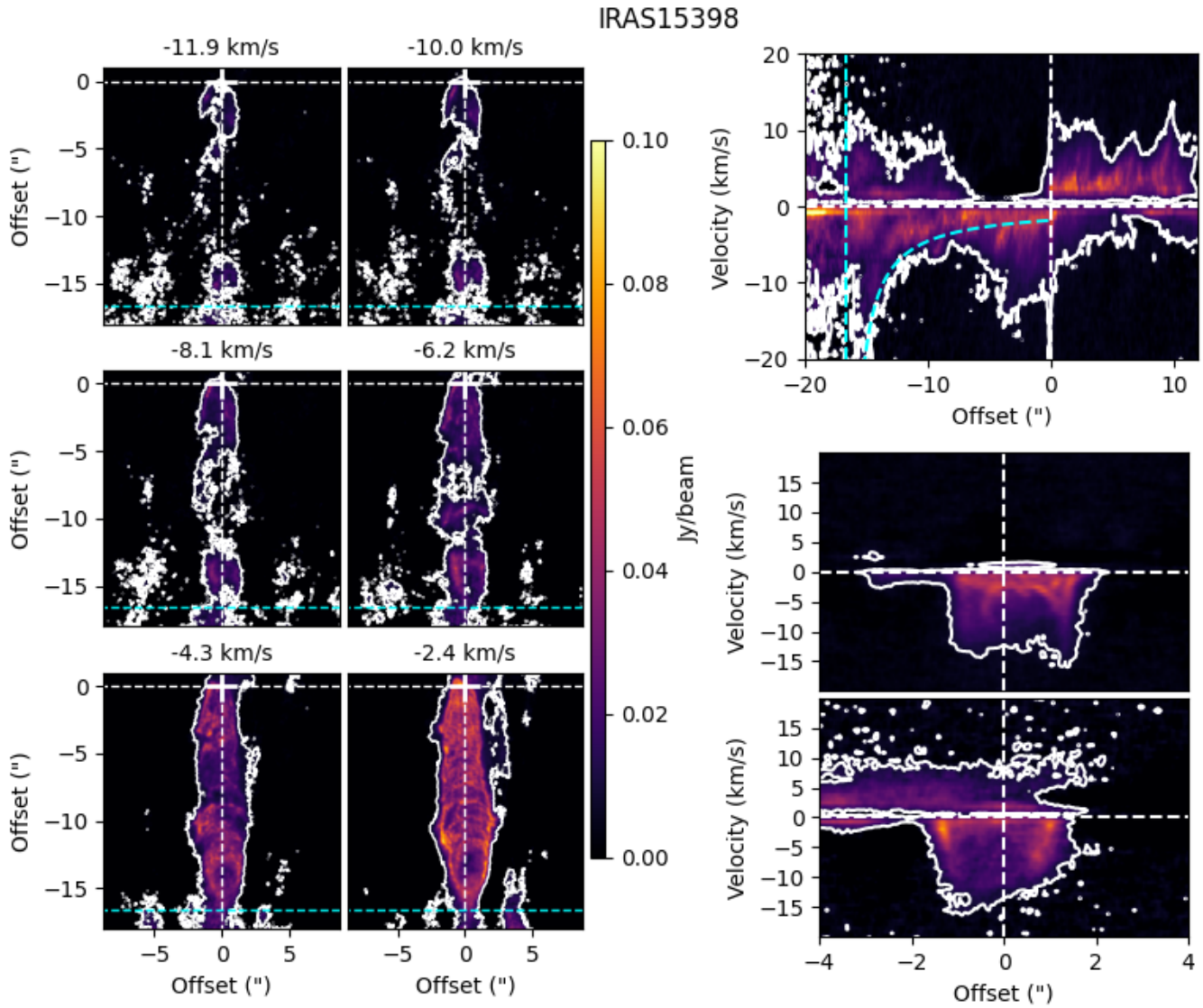


Figure 2. Similar to Figure 1 but for IRAS15398. The position of the bow shock is indicated by the dotted cyan line at offset=16.6″, while the cyan dotted curve in the P-V diagram along the outflow axis (upper-right panel) shows a schematic emission shape as expected from the bow-shock model.

Appendix A, see also Flores et al. 2023). Some other eDisk sources also show significant differences in morphologies between red and blue outflow components, which will be discussed in §4.5 and §5.1.

Other sources where the molecular outflow emission shows characteristics of the wind-driven shell include IRAS 16544-1604, GSS30 IRS 3, R CrA IRAS 32 (IRAS 32 hereinafter), IRAS04166, and TMC-1A. For IRAS04302, some ^{12}CO emission is seen in the blue-shifted outflow on the outflow axis, but it is quite faint and has no distinct shape (Figure 10 in Appendix A). Therefore, it is difficult to identify the type of emission. But the velocity channel map shows an emission which is shifted away from the protostar in the channel maps at higher velocities (see also Lin et al. 2023). We thus possibly classify the outflow of IRAS04302 as a wind-driven shell.

4.2. *Jet-driven bow shock emission*

The molecular outflow emission observed in IRAS15398-3359 (IRAS15398 hereinafter) is more collimated than those of other objects (Figure 2). At the tip of the outflow located $\sim$ 15″ from the source, a high-velocity emission component up to $\sim$ 12 km s^{-1} from the systemic velocity is seen. This relatively high-velocity emission at the tip of outflow observed in ^{12}CO matches the position of a bow shock observed by JWST MIRI (Yang et al. 2022, see also

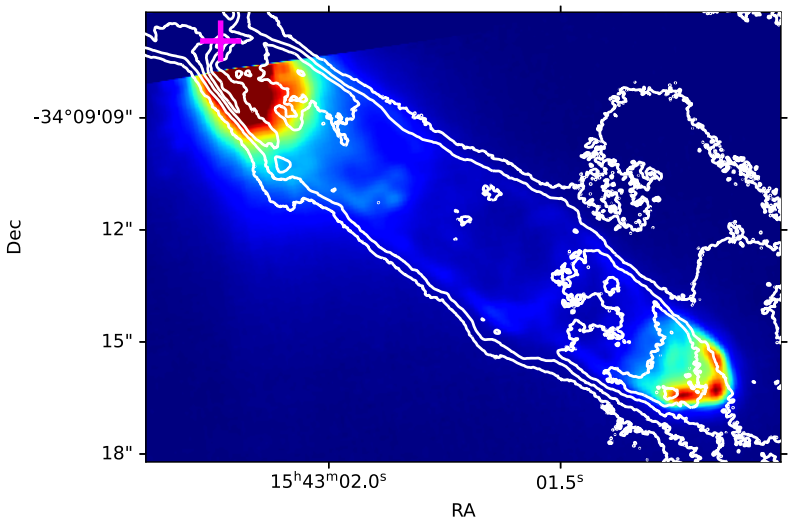


Figure 3. Integrated intensity map of ^{12}CO of IRAS15398 (contours) overlaid on a JWST MIRI image taken with the F1000W filter (colorscale). The position of IRAS15398 is indicated by the magenta cross. The south lobe is the same lobe shown in the channel map in Figure 2.

Okoda in prep)(Figure 3). These features are consistent with the jet-driven bow shock, i.e. a U-shaped collimated outflow with its tip at the position of a bow shock in the jet. In addition, the P-V diagram along the outflow axis (the top right panel in Figure 2) shows a spur-like structure, where the velocity of the emission increases up to the position of the bow shock. This is also consistent with a jet-driven bow shock.

Jet-driven bow shock emission is also seen in the red-shifted outflow of IRAS15398, which also shows a U-shaped collimated shape at velocities of up to ~ 10 km s^{-1} relative to the systemic velocity, at the tip of the outflow emission $\sim 10''$ from the source (Figure 10 in Appendix A). A spur-like structure can also be seen in the P-V diagram peaking at the tip of the emission. All these factors suggest that a bow shock is present in the jet at $\sim 10''$.

With the possible exception of IRAS04169 (see §4.5), none of the other eDisk sources show evidence of jet-driven bow shock emission, suggesting perhaps that it is a relatively rare form of molecular outflow emission at radial distances of $\lesssim$ a few thousands au from the central protostar.

4.3. *Slow disk wind emission*

Slow disk wind emission is the most common form of molecular outflow emission seen in the eDisk objects, with detections in at least 11 of the 19 objects, or out of the 15 sources where molecular outflow emission is detected. A typical example of slow disk wind emission is seen in the blue-shifted outflow of BHR71 IRS 1 (Fig. 4). The velocity channel maps show a conical structure, with the opening angle of the cone decreasing at higher velocities (see also Figure 10 in Appendix A). The P-V diagrams perpendicular to the outflow axis (the right bottom panels in Fig. 4) show that the velocity of the emission decreases with distance from the outflow axis. These properties reflect the onion-like velocity structure of the slow disk wind, where material closer to the outflow axis is moving at higher velocities than material further away from the axis (e.g. de Valon et al. 2020; Pascucci et al. 2023). It indicates that the slow disk wind originates at a range of disk radii in the circumstellar disk of the source.

As mentioned in 3.3, the appearance of a slow disk wind depends strongly on the inclination of the source. While the disk of BHR71 IRS 1 is inclined relative to the plane of the sky ($i = 39°$), that of L1527 IRS is nearly edge-on (van't Hoff et al. 2023). As a result, its appearance is somewhat different, as shown in Fig. 5. Similar to BHR71 IRS 1, the velocity channel maps show a conical structure with the opening angle decreasing at larger velocities (see also Figure 10 in Appendix A). However, the velocity of the emission is lower than the inclined case, with the maximum velocity observed only ~ 5 km s^{-1} relative to the systemic velocity. In the P-V diagrams perpendicular to the outflow axis (the right bottom panels in Fig. 5), the emission traces an elliptical structure which is symmetric about the origin, with the ellipse being larger in P-V cuts at distances further from the source. We note that at this peculiar inclination, i.e. nearly edge-on angle, the P-V diagrams of L1527 IRS appear similar to those of the Type of wind-driven shell aligned

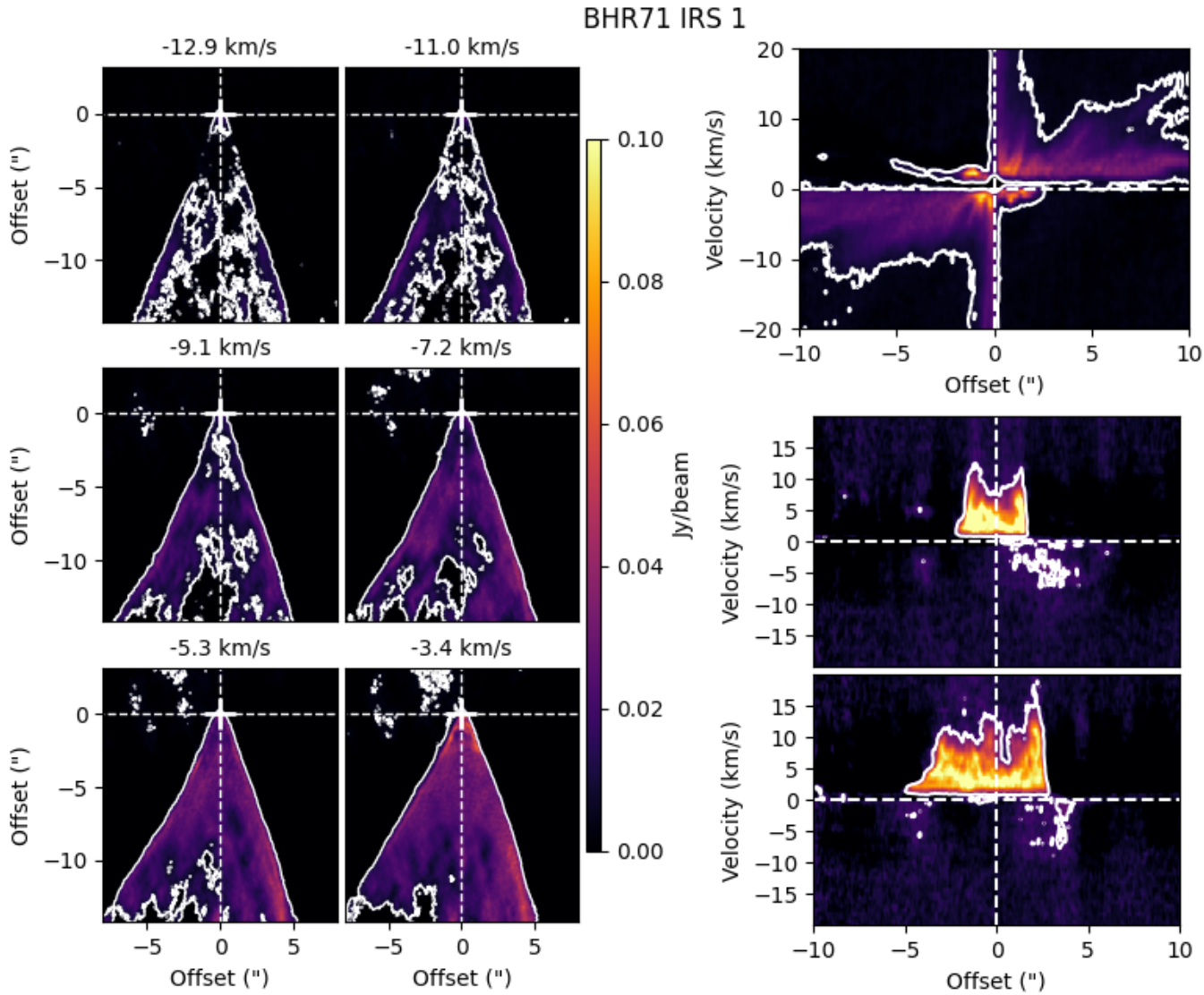


Figure 4. Similar to Figure 1 but for BHR71 IRS1.

to the outflow axis. We can discriminate the slow disk wind and wind-driven shell by combining the channel map and the P-V diagrams.

4.4. *Combination of slow disk wind and wind-driven shell emissions*

When categorizing the observed features, it is important to remember that multiple types of outflow emission could also be present in the same object. For example, wind-driven shells could be present alongside slow disk wind. This is seen in GSS30 IRS3, IRAS04166, IRAS 32, and TMC-1A.

The velocity channel maps and the P-V diagrams parallel to and perpendicular to the outflow axis for the red-shifted outflow of IRAS04166 are shown in Fig. 6 as an example. The channel maps at low velocities show that the outflow has a conical shape that is typical of a slow disk wind. However, at the velocity of $|v - v_{\rm sys}| \sim 9$ km/s, the tip of the cone, which can be regarded as a shell-like structure, is shifted relative to the protostellar position. The shell-like structure is shifted farther away from the protostar at higher velocities [1]. This is typical of a wind-driven shell. Evidence of disk wind emission and shell emission is also seen in the P-V diagrams. The P-V diagram parallel to the outflow axis

[1] The cyan lines in Figure 6 show the best-fit shell model described in §5.1. We fitted the shell at high-velocity channels ($|v - v_{\rm sys}| \gtrsim 9$ km/s), while the cyan line is also plotted in low-velocity channels.

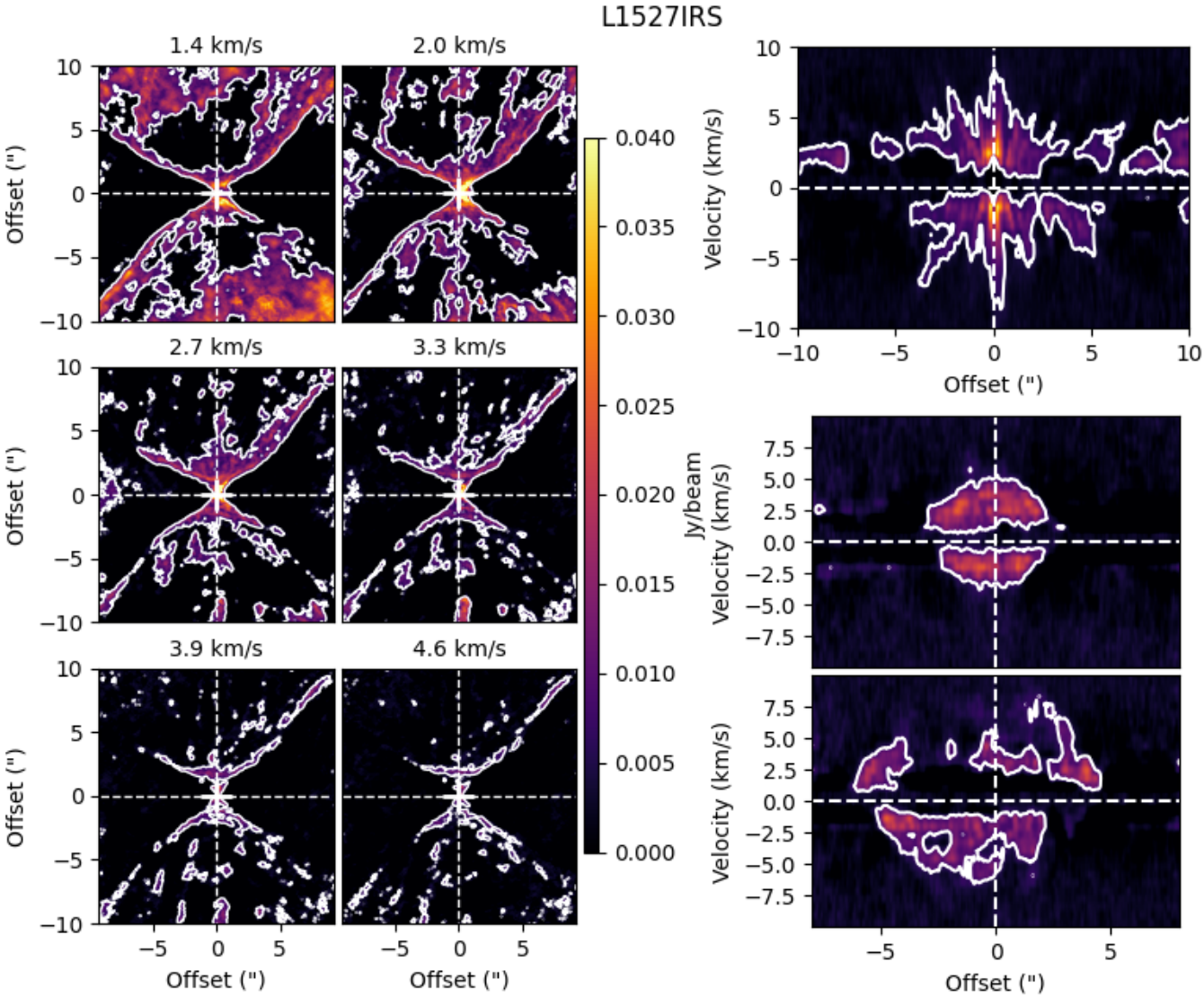


Figure 5. Similar to Figure 1 but for L1527IRS.

(right top panel) shows a parabolic shape in the red-shifted outflow emission, which is evidence of shell emission, while the P-V diagrams perpendicular to the outflow axis (right bottom panels) show the velocity of the emission decreasing with distance from the outflow axis, characteristic of a disk wind, as well as a possible elliptical rim with an inner void in the emission, evidence of a wind-driven shell.

Given that wind-driven shells are typically the result of a wide-angle wind blowing into an ambient medium, it is not surprising to see shell and disk wind emissions present in the same object. This could imply that the disk wind can interact with ambient material to drive shell structures. This statement then raises a question: why do we not see the disk wind feature in other objects showing wind-driven shell emission? In other words, what is the wide-opening angle flow driving the shell? Atomic gas flow would be a candidate. While we see only molecular gas in the millimeter observations, the centrifugal wind can consist of atomic or molecular gas depending on the physical conditions in the launching regions. In this regard, the correlation between the features in mid-infrared emission and our ^{12}CO data in IRAS 15398 is insightful (see also Harsono et al. 2023; Tychoniec et al. 2024).

4.5. *Peculiar features in the outflow of IRAS04169*

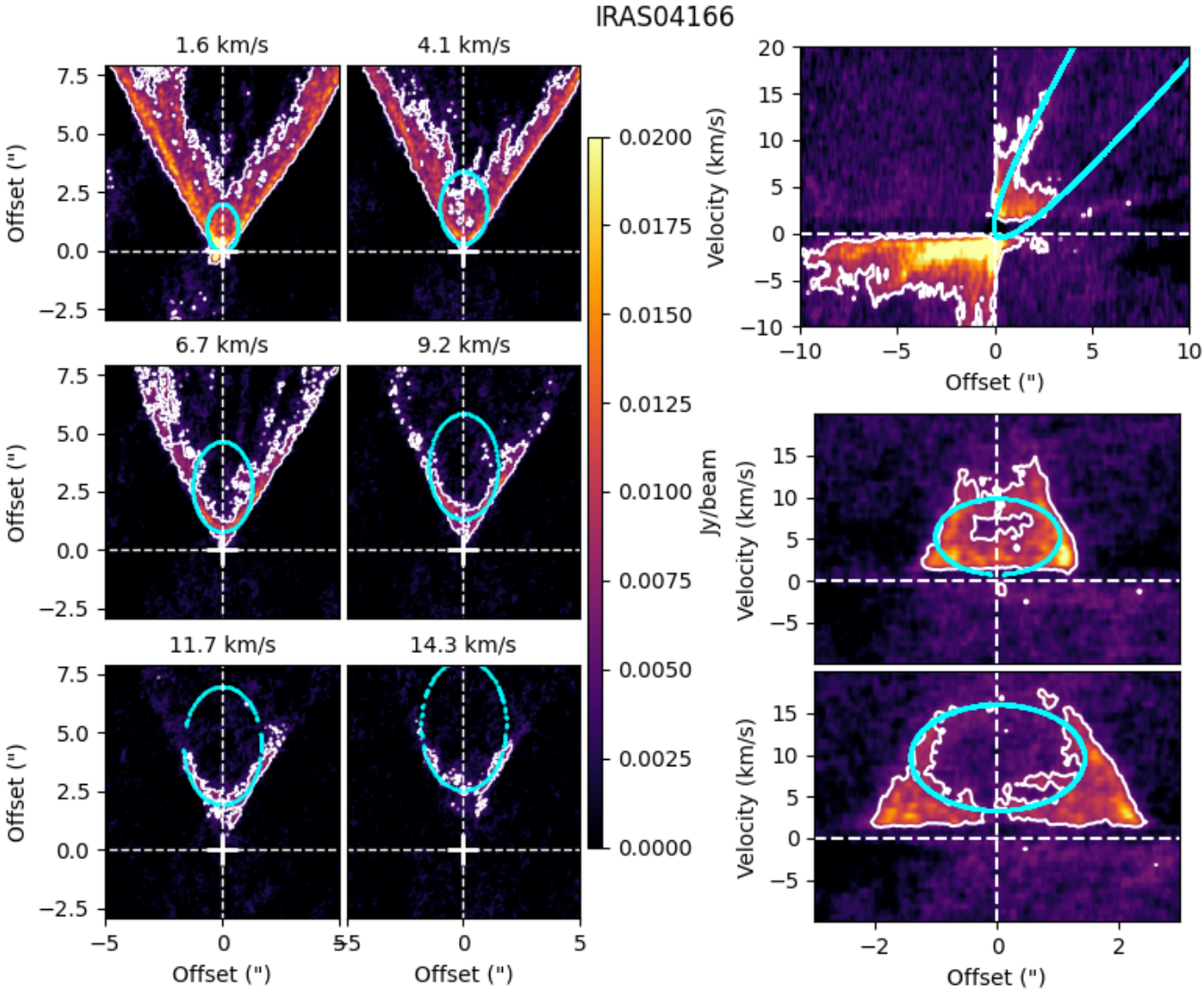


Figure 6. Similar to Figure 1 but for IRAS04166.

When classifying the type of emission observed in the molecular outflow of each object, most of the sources can be classified as one or more of the three types of emission discussed in Section 3. However, the type of outflow emission observed in IRAS04169 is more ambiguous.

While the blue-shifted outflow shows a slow disk wind feature with a parabolic shape in the velocity channel maps (see Figure 10 in Appendix A) and a cone shape in the P-V diagram perpendicular to the outflow, the emission in the red-shifted outflow exhibits a more complex structure. The red-shifted outflow shows some characteristics of a slow disk wind structure; the channel maps show a parabolic structure which becomes narrower at higher velocities (the left panel in Figure 7). The P-V diagram perpendicular to the outflow axis (the right bottom panel) shows a cone shape centered at offset= 0. However, the velocity channel maps (the left panel) also show parabolic shell-like structures offset from the protostar and outflow axis, most clearly at velocities of $v - v_{\rm sys} \sim 2 - 5$ km/s. In the channel maps at $v - v_{\rm sys} = 6 - 9$ km/s , we also see a shell structure at $\sim 8''$ offset from the protostar and $\sim 3''$ offset from the outflow axis. These shells do not show a clear Hubble-law velocity structure, as their distances from the protostar do not significantly change between channels, probably indicating that they are not wind-driven shells. They could be bow shocks, as emission in a shock is expected to exhibit a wide range of velocities. But the curvatures of the observed shells are inverse compared with that we expect for bow shocks driven by a fast flow from the central protostar.

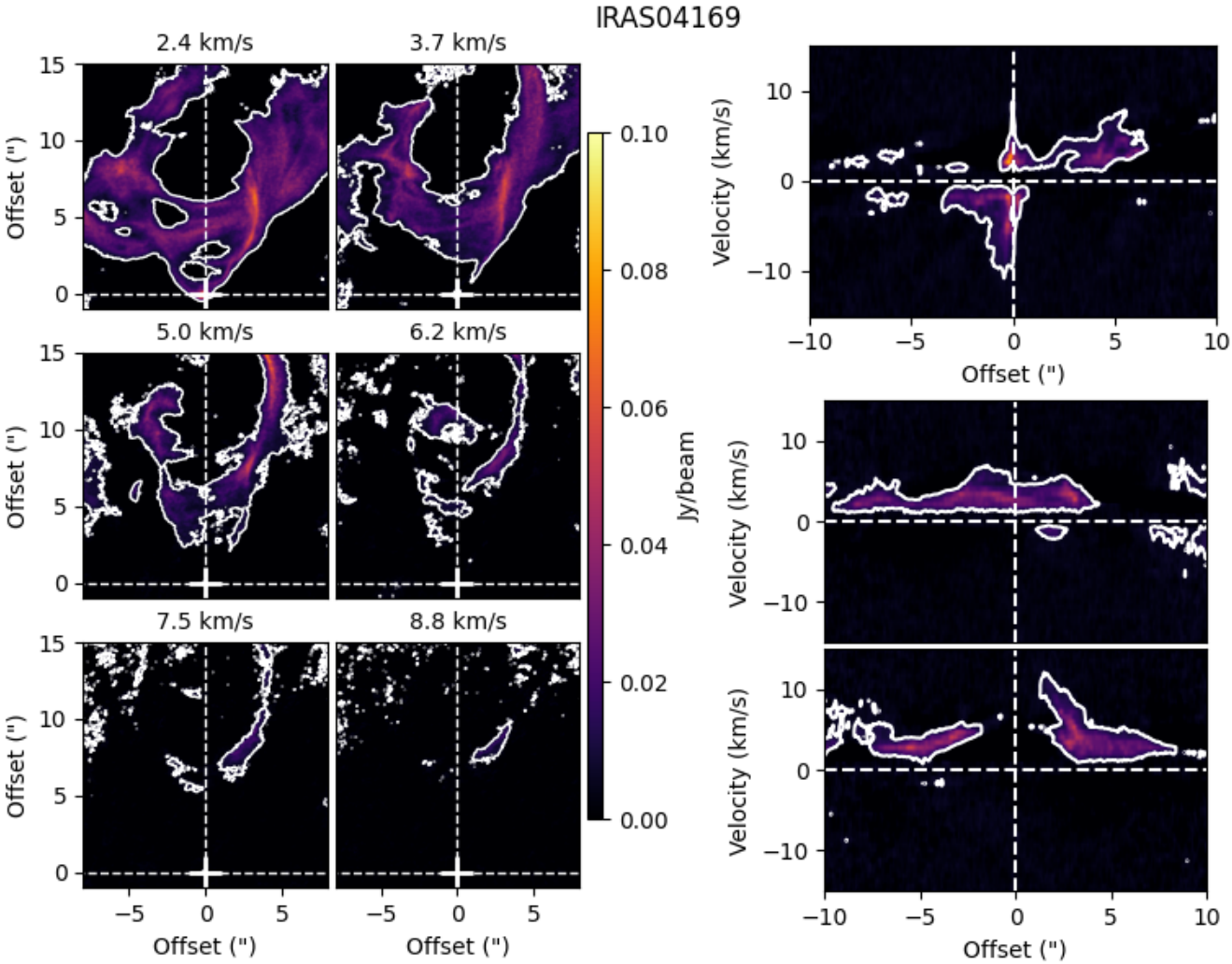


Figure 7. Similar to Fig. 1 but for IRAS04169.

5. DISCUSSION

5.1. *Properties of wind-driven shells*

In this section, we discuss the morphological and kinematic properties of the outflows categorized as wind-driven shells using conventional approaches in the literature (e.g. Lee et al. 2000; Zhang et al. 2019).

5.1.1. *Comparison with a parametric model*

For wind-driven shells there is a simple parametric model introduced by Lee et al. (2000), which allows us to quantitatively discuss their physical properties such as the dynamical age of the shells (e.g. Zhang et al. 2019):

$$\left(\frac{z}{R_0}\right) = \left(\frac{R}{R_0}\right)^2, \qquad v_Z = \frac{z}{t_0}, \qquad v_R = \frac{R}{t_0}. \tag{1}$$

The z-axis is along the outflow axis and the R-axis is perpendicular to the z-axis. The parameter R_0 is the characteristic radius which determines the width of the outflow and is the radius at which $z = R$. The parameter t_0 is the dynamic age and determines the velocity structure of the shell. For each shell we searched for the best values of the parameters R_0, t_0, and i that can explain the observations by plotting the corresponding model on top of the velocity channel maps and P-V diagrams to compare by eye. The best parameters are summarized for each source in Table 5, while the model shells overlaid on the observed channel maps and the P-V diagrams are shown in Appendix B (Figure 11).

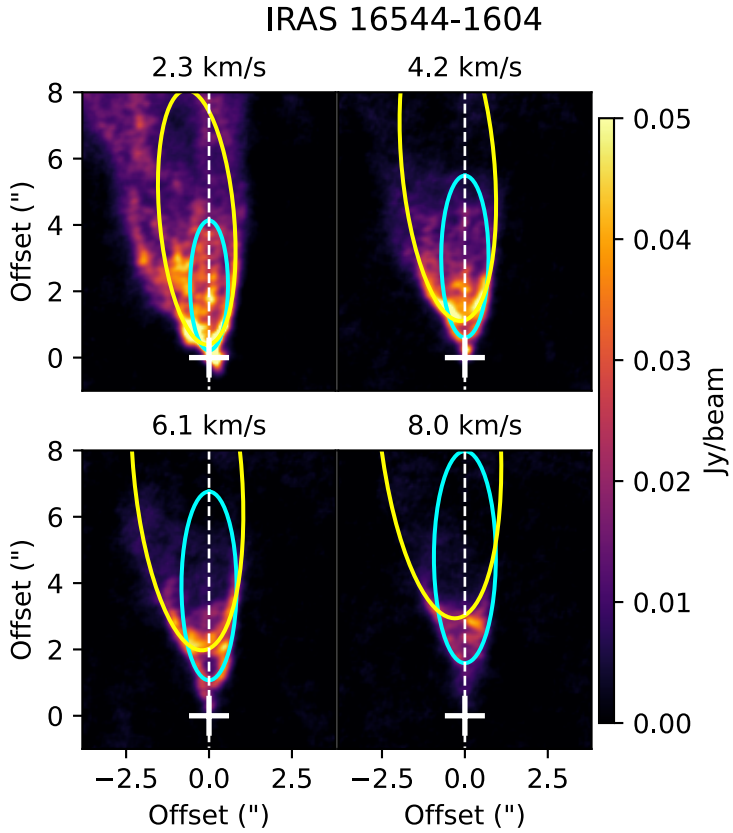


Figure 8. The velocity channel map of IRAS 16544-1604 (color scale) with the model shells R1 (cyan) and R2 (yellow) fitted to the emission shown by the ellipses.

We note that the actual shell structure depends on the density and velocity distributions of the driving wind and ambient gas (Shang et al. 2006, 2020). The agreement between our observations and this simple model thus varies among objects. For some objects, the model reasonably agrees with the P-V diagram, but a deviation is apparent in the channel map, and vice versa. When the shell structure is not clear in the channel map, but a Hubble flow or an ellipse is more clear in the P-V diagram, we prioritize the comparison of the model with the P-V diagram, rather than with the channel map.

For Oph IRS 63, the representative object of wind-driven shell emission, the cyan lines in Figure 1 depict the best model. Assuming a distance to the source of 132 ± 6 pc (Zucker et al. 2020), we find that the shell structure was best described by a model with $R_0 = 0.6''$, $t_0 \sim 380$ yrs, and $i \sim 47°$ (B1 in Table 5). This inclination is in agreement with the value of $i = 46.7°$ derived from the Gaussian fit to the continuum disk emission by Ohashi et al. (2023) and Flores et al. (2023).

The channel map of Oph IRS 63 shows another shell, which is much wider than the one described above (see Figure 11 in Appendix B). Its best parameters are $R_0 = 6''$ and dynamic age $t_0 = 1600$ yr (B2 in Table 5). However, the agreement is much worse compared with the narrower shell. While the model can explain the shell emission in the channel map at lower velocities ($|v - v_{\rm sys}| \lesssim 8$ km/s), at higher velocities the observed shell emission appears to be "slower" than the model shell. Such deviation is also found for other larger, older shells. This may be explained by the deceleration of the flow due to interaction with ambient material, which is not considered in the simple model of equations (1). The deceleration would have a greater impact on higher-velocity material than the lower-velocity material, explaining the difference between the model and observed shells at high-velocity channels. It is also important to note that this deceleration indicates that the dynamic ages derived for the shells are upper limits on the age of the shell rather than their true age (Zhang et al. 2019).

5.1.2. *Dynamical timescales*

The dynamical timescales obtained above infer the time variability of the outflow. The mass-loss rate in protostellar outflows is known to be highly variable. This is most notably seen in the form of a series of emission knots in the jets of YSOs, which are believed to trace bow shocks caused by variations in the velocity and direction of the flow (e.g. Zinnecker et al. 1998; Plunkett et al. 2015; Lee et al. 2018, 2024). While evidence of variability has not been seen as frequently in the case of molecular outflows, Zhang et al. (2019) found multiple wide-angle shell structures in the CO emission of the HH46/47 molecular outflow. They argue that these shell structures are the result of entrainment of ambient material around the source by a series of outbursts from the central object. Our eDisk data shows such evidence of variability in the mass-loss rate of the outflows in the form of the wind-driven shell emission in 7 sources. The difference between the dynamical ages of the shells in the same source, i.e. the timescale of variability, is found to be several hundred yrs (Tables 5). In the case of Oph IRS63, for example, the observed shells in the blue-shifted outflow indicate that outbursts occurred in this object ~ 380 and ~ 1300 years ago.

Future studies of shell structures within outflows could benefit from comparing the dynamic ages of shells to the dynamic ages of emission knots observed in collimated jets of the same source. This could tell us if the wide-angle wind that drives the shell is related to the jet. The dynamical ages could also be compared with the dates of accretion outbursts detected in the source, e.g. FUor or EXor outbursts, if any, given that the accretion and mass loss processes in star formation are closely linked. Kim et al. (2024), for example, recently observed B335 and found that the dynamical age of the high-velocity ^{12}CO outflow is similar to that of the accretion burst probed by the mid-infrared brightness.

Table 5. Shell Model fits for the eDisk sources with detected wind-driven shell emission.

Source	$i_{\rm cont}$	PA$_{\rm cont}$	Shell	t_0	R_0	i	PA
		yr	arcsec	$^\circ$	$^\circ$		
IRAS 16544-1604	73	45	B1	2500	4	73	128
			R1	100	0.2	73	335
			R2	190	0.4	73	340
GSS30 IRS3	64	109	B1	1400	18	82	18
			R1	40	0.7	85	198
			R2	400	4	85	198
IRAS04166	47	122	R1	250	0.5	49	212
R CrA IRAS 32	69	135	B1	1200	4	69	43
			R1	850	3.5	69	223
IRAS04302	84	175	B1	> 200	> 0.3	84	85
TMC-1A	52	76	B1	400	0.5	52	346
			B2	800	1.0	52	346
Oph IRS63	47	149	B1	380	0.6	47	240
			B2	1300	5	47	240
			R1	1600	6	47	90

As noted in §4, the properties of blue-shifted and red-shifted outflows differ in some objects. Specifically, Table 5 indicates that the dynamical ages of the shells are different between the red- and blue-shifted outflows. In the case of IRAS 16544-1604, for example, the blue-shifted outflow has one shell of dynamical age of 2500 yrs, while the red-shifted component has two shells of dynamical age of 100 yr and 190 yr (R1 and R2 in Figure 8), These differences may be due to different properties of the ambient medium on either side of the outflow. The swept-up shells could be too faint to be detected if the density of the ambient material is low. Density distribution of the ambient material also affect the circumstellar magnetic fields, which plays essential role in driving winds.

5.1.3. *Position angle and inclination*

Other parameters of the shell model (Eq. 1) can also yield interesting information on the geometry of the outflows, i.e. their position angles (PA) to be compared with that of the dust continuum, i.e. PA$_{\rm cont}$. Among our targets, IRAS 16544-1604 and Oph IRS 63 are categorized as wind-driven shells without the indication of coexistence with a slow disk wind. It is interesting that the PA of their outflows (i.e. shells) are not perpendicular to that of the

dust continuum. For example, the PAs of the red-shifted shells in IRAS 16544-1604 are 335 °and 340 °, while the PA of its dust continuum is 45 °(Table 4, see also Kido et al. 2023); the latter values should be 65-70 °if the disk is perpendicular to the outflow. For this source, the dust continuum is considered to be tracing disk. Kido et al. (2023) analyzed $C^{18}O$ (2-1) emission to find that the gas rotation in the continuum emitting region is close to the Keplerian rotation. Similarly Kepler-like rotation is confirmed around Oph IRS 63 (Flores et al. 2023).

Interestingly, the PAs of the two red-shifted shells of IRAS 16544-1604 also differ by $\sim 5°$, suggesting that the outflow axis changed between the two outbursts that created the shells, while the dynamical ages of the two shells differ only by 90 yrs (Figure 8). We note that temporal changes in the outflow axis have been found and discussed in another eDisk target IRAS 15398; in addition to the outflow shown in Figure 2, there are two sets of old outflow candidates (with the dynamical timescale of $10^3 - 10^4$ yr) misaligned by $20 - 90°$ (Okoda et al. 2021; Thieme et al. 2023; Sai et al. 2024). Such temporal variation of the outflow axis could be due to the accretion of gas with various angular momentum vectors, which could change the angular momentum axis of the star-disk system. The impact should be more significant for lower mass protostars, in which the ratio of newly added momentum to the total momentum should be relatively large. Indeed, the protostellar mass of IRAS 16544-1604 and IRAS 15398 are estimated to be relatively small ($\sim 0.1 M_\odot$) among eDisk objects (Okoda et al. 2021; Thieme et al. 2023; Kido et al. 2023).

From the comparison with the simple shell model, the inclination angle i of an outflow is also estimated, and it can be compared with the disk inclination ($i_{\rm cont}$ in Table 1) that is estimated from the major and minor axis of the dust continuum emission. While these angles are similar to each other for most of the sources, for GSS30 IRSS 3 we obtained $i \sim 82$ and $\sim 85°$ for the blue and red-shifted shells respectively, compared to $i_{\rm cont} = 64.3 \pm 1.5°$ (Santamaria-Miranda et al. 2024). However, given that the disk may be geometrically thick, $i_{\rm cont}$ is only a lower limit of the disk inclination. The inclination angles of the disk and the outflow thus could be consistent, and the disk could be nearly edge-on.

For IRAS04302, it is difficult to derive a shell model due to the faint nature of the emission and lack of distinct elliptical shape in the channel maps. The value of R_0 is not very well constrained, and a wide range of values of R_0, t_0 and i can be consistent with the data. Assuming that the inclination angle of the disk and outflow is the same, we get a lower limit for R_0 and t_0 based on the smallest shell that can fit the data.

5.2. *Objects with no outflow detected*

While most of the sources show a certain type of molecular outflow emission, four sources, Ced 110 IRS 4, IRS 5N, IRS 7B, and IRS43, show no clear ^{12}CO outflow in our data. An important question to address is whether this is because the emission has been resolved out or because there is indeed no outflow present.

^{12}CO molecular outflows have previously been detected at larger scales (Yıldız et al. 2015; van Kempen et al. 2009; Bontemps et al. 1996) for Ced 110 IRS 4 and IRS 7B. So, it is reasonable to assume that the ^{12}CO emission is simply resolved out in the eDisk observations as the spatial scale of these previous studies is larger than our maximum resolvable scale of $2'' - 3''$. Observations obtained with ALMA in short-baseline configurations (Plunkett et al., in prep.) will help confirm this.

As for IRS43, Bontemps et al. (1996) observed a velocity gradient of the ^{12}CO (2-1) emission in the east-to-west direction using IRAM 30 m and interpreted it as an outflow. This velocity gradient, however, is consistent with the envelope rotation found in Narayanan et al. (2023) using ^{12}CO, ^{13}CO, and $C^{18}O$ (2-1) data of eDisk. Narayanan et al. (2023) also showed that the axis of the outflow cavity runs from north to south, even though outflow emission is not clearly detected in this direction. This example shows having both large-scale and high-resolution observations is crucial for identifying outflows.

As for IRS 5N, previous studies (e.g Bontemps et al. 1996; Yang et al. 2018) failed to obtain any clear molecular outflow detection, and there has been no clear detection of other jet/outflow tracers for this source. While cm-wavelength radio emission was detected by Miettinen et al. (2008), which could at least partially be thermal free-free emission from an ionized base of the jet, the negative spectral index and rapid variability of the emission suggest it is mainly gyrosynchrotron from the stellar magnetosphere. IRS 5N was also suggested as the possible source of a H_2 shock detected by Kumar et al. (2011). The high density of sources in the R CrA region, however, makes it difficult to be certain whether it is associated with IRS 5N. Based on the lack of detection in this and other studies, it is tempting to speculate that IRS 5N lacks an outflow or has a very weak outflow, although the complex spatial and kinematic structures of the R CrA region due to the high density of sources could hamper the clear detection. In theoretical studies, it has been proposed, for example, that a misalignment between the cloud rotation axis and the magnetic field could suppress outflow activities (Hirano et al. 2020; Takaishi et al. 2024).

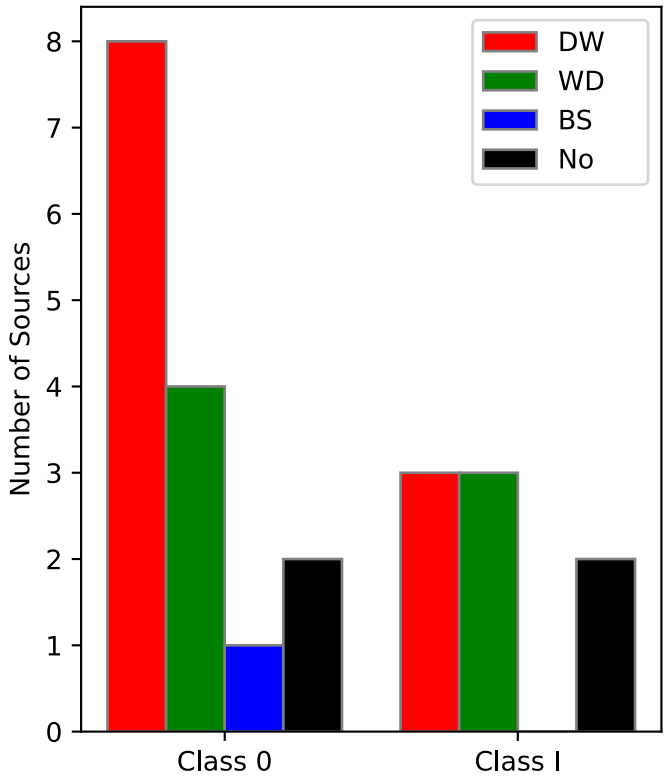


Figure 9. A bar chart showing the number of sources with wind-driven shell (WD), disk wind (DW), and bow shock (BS) emission in Class 0 and I. The black bars labeled "No" depict the number of sources without clear outflow emission.

It is also worth noting that Ced 110 IRS 4, IRS 7B, and IRS 43 are three of the four close-binary systems with the apparent separation $\lesssim$ 250 au on the plane of the sky in the eDisk targets. It indicates that the existence of a close binary may have an effect on the structure of the molecular outflow on small scales. Another close-binary system in eDisk targets, IRAS 32, however show clear outfow emission categorized as the combination of slow disk wind and wind-driven shells, which we assume are launched from the primary (IRAS 32 A).

5.3. *Is there a correlation between the class of the source and the outflow features?*

An important question is whether there is a correlation between the type of outflow emission observed and the evolutionary state of the central protostar. Naively, one might expect the wind-driven shell to be more common in Class 0 objects as there should be more ambient material present than in Class I. Our eDisk data, however, does not immediately support this hypothesis: Figure 9 summarizes the number of sources with wind-driven shell, slow disk wind, bow shock, and no clear outflow emission for each class. Differences between Class 0 and I are not clear. While 4 out of the 12 Class 0 sources (33%) show wind-driven shells, 3 out of the 7 Class I sources (43%) also show such shells. Slow disk wind emission is common in both Class 0 (67%) and Class I sources (43%). Data of larger samples and with a larger maximum recoverable scale (e.g. combination of 12m-array with ACA) would be desirable for further studies.

6. CONCLUSIONS

We have analyzed outflow features observed with ^{12}CO (2-1) towards 19 protostellar sources covered in eDisk. Various emission features are detected towards 15 objects. We categorized the outflow emission of the 15 objects into three types, wind-driven shell, bow shock, and slow disk wind, based on the specific features found in the velocity

channel maps and the P-V diagrams along and perpendicular to the outflow axis. Seven objects are categorized as wind-driven shell, 1 object as bow shock, and 11 objects as slow disk wind, among which 4 objects show both slow disk wind and wind-driven shell. For the bow shock emission of IRAS15398, the shell structures found in our ^{12}CO data correlate well with a bow shock feature observed with JWST. We do not find any clear correlation between outflow features with the evolutionary stage, i.e. Class 0 and I.

Some objects show different features in their blue- and red-shifted outflows. For example, GSS 30 IRS3, IRAS04166, and TMC-1A show mixed features of disk wind and wind-driven shell only in blue or red components. The blue-shifted outflow of IRAS04169 is categorized as disk wind, while its red-shifted flow shows a peculiar feature that is difficult to categorize. The blue-shifted and red-shifted outflows are not aligned with each other in IRAS 16544-1604, IRAS 15398, IRAS 04169, L1489, and Oph IRS63. Misalignment between the outflow axis and the disk minor axis, which is determined by 2D Gaussian fitting of the dust continuum emission, is also found for several objects (e.g. L1489 IRS and IRAS 16544-1604).

We adopted the model of Lee et al. (2000) and compared it to the emission categorized as wind-driven shell to derive e.g. dynamical ages of the shells. The derived dynamical ages are different between the shells in the blue- and red-shifted outflows, which is probably caused by the difference in the amount of ambient material with which the wind interacts. At high velocity channels, old shells show signatures of deceleration (e.g. $|v - v_{\rm sys}| \gtrsim 8$ km s^{-1} in the blue-shifted wide shell of Oph IRS 63). In IRAS 16544-1604, two shells with dynamical age of 100 yr and 190 yr are misaligned, indicating the variation of the outflow axis in a short timescale.

In our eDisk samples, four objects, Ced110 IRS4, R CrA IRS 7B, R CrA IRS 5N and Oph IRS43, do not show outflow emission. Among them, large-scale outflows that could be resolved out in eDisk data are previously observed towards Ced110 IRS4 and R CrA IRS 7B. As for Oph IRS43, the large scale CO emission previously considered as outflow (Bontemps et al. 1996) turned out to be a rotating envelope component described in the eDisk first-look paper (Narayanan et al. 2023). These results imply that it is critical to sample both large- and fine-scale structures for unambiguous identification of molecular outflows.

While we adopted a simple categorization of outflows and a simple parametric model of wind-driven shell for comparison with the observation, we do not aim to exclude any models or physical mechanisms not explored in the present work. Our motivation is to report a variety of structures and features, some of which are previously observed individually but now with a group of sources with a uniform data quality. Further observational and theoretical studies are necessary to understand this variety. For example, it is important to compare our results with observations of the eDisk sample at shorter baselines. This will allow the detection of large-scale molecular emission in the outflows that may have been resolved out in the data presented here, particularly for those objects where no molecular outflow emission was detected. Such observations have already been carried out and will be presented in a future publication.

APPENDIX

A. CHANNEL MAPS

Figure 10 shows the velocity channel map of BHR71 IRS2. The complete figure set for all the eDisk sources (19 images) is available in the online journal. The maps are oriented so that the axis of the blue-shifted outflow points upwards. The outflow axis is set to be perpendicular to the major axis of the dust continuum emission for most of the objects. For some objects, however, the outflow is significantly misaligned to that direction. For those objects the outflow axis is determined by finding the axis of symmetry of the outflow by eye, or by fitting the data with a simple shell model in the cases of objects with wind-driven shells (§5.1). The velocities are relative to the systemic velocity. Table 4 summarizes the systemic velocity and position angle adopted in drawing the channel maps.

Fig. Set 1. Velocity channel maps

B. MODEL FITTING OF WIND-DRIVEN SHELLS

Figure 11 shows (a) the velocity channel map, (b) the P-V diagram along the outflow axis, and (c) the P-V diagram perpendicular to the outflow axis of the blue-shifted outflow of IRAS 16544-1604. The feature categorized as wind-driven shell is fitted with a simple model of Eq. 1 in §5.1. The complete figure set (11 images) is available in the online journal.

Fig. Set 2. Model fitting of wind-driven shells